Feasibility Review: Streamflow Prediction at Unmonitored Rivers

Oct 9, 2026 · @Derek

Bottom line

The proposal is **feasible as a course project, but only with a narrower scope**. The data are real and free, the computing needs fit on a laptop, and the 0.40 median NSE target is a reasonable bar. Three things need fixing before work starts:

1. The factorial design is too large for the timeline.
2. The central hypothesis is likely to come out "no" for a structural reason the team should state up front.
3. The validation protocol, which the proposal doesn't yet describe, is where most of the risk to the results sits.

What is sound

The motivating benchmark is cited correctly. Kratzert et al. trained LSTMs on 531 CAMELS basins with k-fold validation, so each prediction came from a basin that supplied no training data. The out-of-sample LSTM had a higher median NSE (0.69) than either the per-basin calibrated SAC-SMA (0.64) or the National Water Model (0.58) (Water Resources Research, 2019).

The data source is appropriate. Caravan (Kratzert et al., Scientific Data, 2023) standardizes and aggregates seven existing large-sample hydrology datasets, with forcing, streamflow and static attributes for 6,830 catchments. CAMELS-GB is among the source datasets. Good candidates for the drought-prone third region are CAMELS-AUS (150 basins in Caravan) and CAMELS-BR (376 basins).

Excluding flow-derived attributes is the right call. Caravan's attributes come mostly from HydroATLAS and ERA5-Land, covering hydrology, physiography, climatology, soils and geology, land cover, and anthropogenic influences. The team should still audit the "hydrology" category for anything computed from discharge.

The research question has real grounding. Oudin et al. (WRR, 2008) compared regionalization approaches on 913 French catchments. In that dense network, spatial proximity worked best, regression was least satisfactory, and physical similarity fell in between; all three lagged far behind full calibration. So "do similarity groupings help?" is a legitimate, still-open question.

Major issues

1. Cluster ID is redundant with full attributes

A k-means or agglomerative cluster label is a deterministic function of the attributes. A tree model that already sees those attributes can, in principle, learn any partition the clustering finds. So "cluster ID beats full attributes" can only win through variance reduction, which matters when there are few training basins or many noisy attributes. It cannot win by adding information.

Frame the hypothesis that way: clustering helps when training basins are scarce. Test it by varying how many training basins the model sees, as Gauch et al. (Environmental Modelling & Software, 2021) did; they found tree- and LSTM-based models similarly accurate on small datasets, with LSTMs superior given more training data.

Physical similarity is also a weak proxy for hydrological similarity. Oudin et al. (WRR, 2010) found the overlap between physically similar and hydrologically similar catchment pools was significant for only 60% of catchments. Evaluate whether clusters are hydrologically coherent, not just geometrically tidy.

2. The ungauged evaluation protocol isn't specified

This is the biggest leakage risk.

- **Hold out whole basins.** Split by basin, not by day.
- **Fit preprocessing on training basins only.** Refit scalers, PCA and clustering inside each fold; otherwise test-basin attributes shape the clusters.
- **Use spatially blocked folds.** Random basin folds put near-neighbours of a test basin in the training set, inflating both the ML scores and the nearest-gauge baseline. Roberts et al. (Ecography, 2017) showed that ignoring such structure seriously underestimates predictive error, and recommend block cross-validation wherever dependence exists.
- **Also split in time.** Train on earlier years and test on later ones so autocorrelated weather can't leak.

3. The design is combinatorially too large

Three regions × two clustering methods × several values of k × four basin-information modes × three model families × 2^3 preprocessing toggles × hyperparameter grids × CV folds comes to thousands of fits. Kernel methods on millions of basin-days will dominate runtime even with the Nyström approximation.

Two items also sit outside an ML course scope: evaluating "additional potential influences beyond the data present in the Caravan dataset," and the synthetic benchmark (issue 5).

4. The forcing data will cap achievable skill

Caravan uses ERA5-Land, which differs from the Daymet, NLDAS and Maurer forcings of the original CAMELS (4TU.ResearchData). In a Frontiers in Water study (2023) — a lower-impact venue, so weight it accordingly — an LSTM trained on ERA5 reached a median daily NSE of 0.54 versus 0.71 with local higher-resolution forcing, though ERA5 did better in most western and northwestern US catchments.

The 0.69 benchmark is therefore not apples-to-apples. The 0.40 target remains plausible but not guaranteed for tabular models, especially in arid basins, where NSE is unstable.

5. The synthetic benchmark risks circularity

If data are generated with parameters drawn per cluster, clustering wins by construction. Describe it as a positive control (does the pipeline recover a known structure?), and specify the generator, for example a simple bucket model with parameters drawn from the attribute space.

Counter-evidence and balance

- **For the team's premise.** Kratzert et al. (2019) concluded that available catchment attributes typically carry enough information about similarities and differences between catchments to support out-of-sample simulation. Attributes matter in the ungauged setting.
- **Against an assumption the team might make.** Li et al. (WRR) showed that replacing physical descriptors with a vector of random values gave comparable performance — but explicitly in a gauged scenario, where attributes can act as mere basin identifiers. Don't generalize gauged-setting feature-importance results to the ungauged case.

- **On spatial vs. physical donors.** Oudin et al. (2008) found spatial proximity beat physical similarity in France's dense network, and suggested combining the two could help. The nearest-gauge baseline may be hard to beat in CAMELS-GB, where basins are close together, and easier in sparser networks.

Minor points

- **Terminology.** The proposal calls the approach "simple tabular unsupervised ML models." The predictive models are supervised; only the clustering is unsupervised.
- **Metrics.** NSE is dominated by high flows. Given the drought motivation, add KGE and a low-flow metric such as NSE on log flow.
- **Log-flow back-transform.** Back-transforming log predictions biases the mean low unless corrected. Evaluate both spaces explicitly.
- **Nearest-gauge transfer.** State the scaling rule (e.g., drainage-area ratio) and the distance metric.
- **Cross-region transfer.** US-to-GB transfer is a separate research question; treat it as optional.

Suggested minimum viable scope

With these changes the project is clearly doable in a semester.

1. **Regions:** CAMELS-US plus one drought-prone Caravan subset (AUS or BR). GB is a stretch goal.
2. **Model:** LightGBM only. Drop kernel regression, or run it on one configuration as a sanity check.
3. **Basin-information conditions:** none, cluster ID, raw attributes, plus a nearest-gauge baseline. One clustering method, k swept over 3–4 values.
4. **Preprocessing:** fix weather windows at 7/30/90/365 days rather than toggling them; log-transform the target.
5. **Validation:** spatially blocked basin k-fold combined with a temporal split, all preprocessing refit per fold.
6. **Key added experiment:** vary the number of training basins (e.g., 25/50/100/all) to find where cluster information starts or stops helping.
7. **Synthetic data:** one small positive-control experiment, if time permits.

A likely, and still useful, finding: cluster ID roughly matches full attributes when training basins are scarce and falls behind as data grows.

Sources

- Kratzert et al. 2019, Water Resources Research — Toward Improved Predictions in Ungauged Basins
- Kratzert et al. 2023, Scientific Data — Caravan
- Caravan v3, 4TU.ResearchData
- Gauch et al. 2021, Environmental Modelling & Software — The Proper Care and Feeding of CAMELS
- Oudin et al. 2008, WRR — Spatial proximity, physical similarity, regression and ungaged catchments
- Oudin et al. 2010, WRR — Are seemingly physically similar catchments truly hydrologically similar?
- Li et al., WRR — Regionalization: From Physical Descriptors to Random Vectors
- Roberts et al. 2017, Ecography — Cross-validation strategies for structured data
- Frontiers in Water 2023 — Predicting streamflow with LSTM networks using global datasets